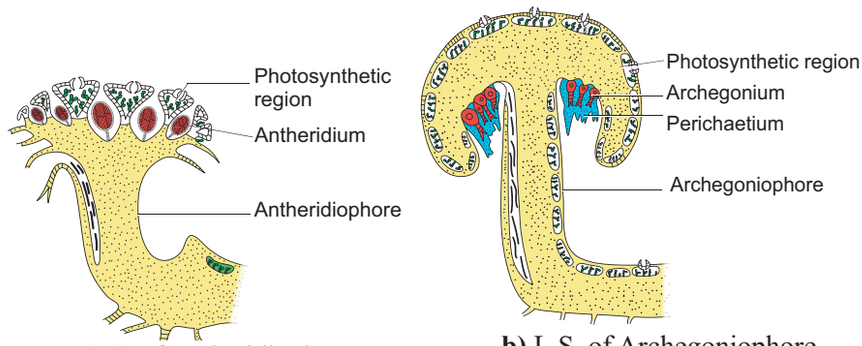
The floor of the chambers bears simple or branched green filaments. Cells of filaments are involved in photosynthesis. The photosynthetic region is followed by storage region. It is made up of several parenchymatous cells arranged without intercellular spaces. The cells of this region contain starch grains and protein granules. The lower epidermis possesses rhizoids and multicellular scales.



a) L.S. of Antheridiophore **b)** L.S. of Archegoniophore

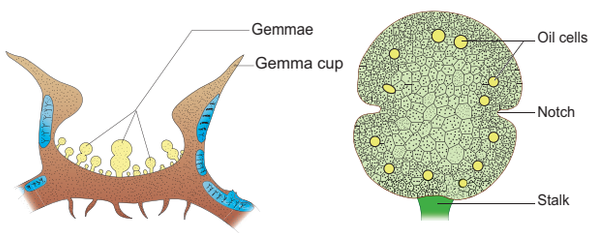
Figure 2.12: *Marchantia* - Sex organs

Reproduction

Marchantia reproduces by vegetative and sexual methods.

Vegetative reproduction

Vegetative reproduction takes place by progressive death and decay of thallus, formation of adventitious branches and by germination of gemmae. Death and decay of the thallus starts from posterior end. When it reach the point of dichotomy, two apical parts of the thallus get separated. Each one develops into an independent thallus. Adventitious branches are produced on the ventral surface of the gametophyte. The branches get separated from the parent thallus and grow into independent gametophytes. Gemmae are specialized multicellular asexual reproductive bodies. They are formed in small cupules known as gemma cups, present on the dorsal surface of the thallus. Usually the gemmae present on the male thallus form male plants and those on the female thallus give rise to female plants (Figure 2.11).



a) L.S. of Gemma cup **b)** A gemma enlarged

Figure 2.11: Vegetative reproduction in *Marchantia*

Sexual reproduction:

In *Marchantia*, sex organs are borne on special stalked receptacles called the gametophores. Those bearing antheridia are called antheridiophores and archegonia bearing structures are called archegoniophores (Figure 2.12). *Marchantia* is heterothallic or dioecious. i.e., male and female receptacles are present on different thalli. The sex organs in bryophytes are multicellular. The Male sex organ is called antheridium. It produce biflagellate antherozoids. The Female sex organ is flask shaped called archegonium and produces a single egg. Water is essential for fertilization. The antherozoids are released into water and are attracted towards archegonium through chemotaxis. Although many antherozoids enter the archegonium, only one fuses with the egg to form zygote. The zygote represents the first cell of the sporophytic generation. Zygote develops in to a multicellular structure called sporophyte. (Figure 2.13).

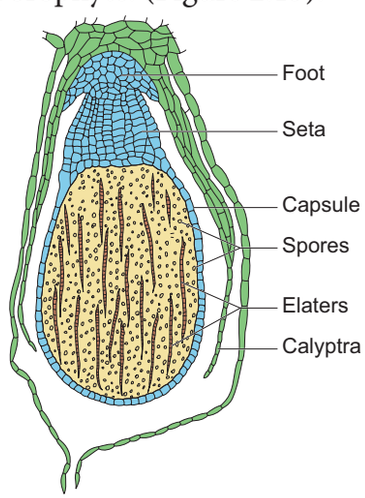


Figure 2.13: *Marchantia* - L.S. of Sporophyte

11th Botany_EM_Chapter 2.indd 44 06/01/2023 19:41:26